WHATSAPP CHAT SENTIMENT ANALYZER

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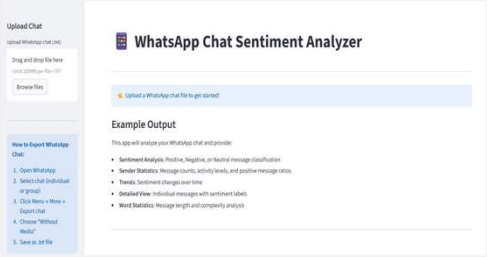
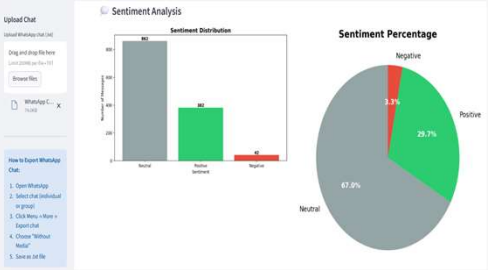
INTRODUCTION

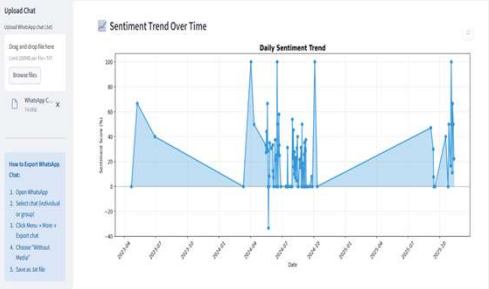
WhatsApp Chat Sentiment Analyzer is an NLP-based tool for emotion analysis of WhatsApp conversations. It requires users to upload their exported chat file, which the system cleans and preprocesses using NLTK techniques, such as tokenisation, stopword removal, and text normalisation. Finally, after processing, the system uses a learning-based model, such as LSTM, to classify each message into positive, negative, or neutral. Results are then showcased using graphs and charts, allowing one to clearly understand the emotional patterns, mood shifts, and communication trends in the chat.

OBJECTIVES

•To develop an NLP-based system that analyses the sentiment of WhatsApp chat messages.
•To preprocess chat data using tokenisation, stopword removal, and text normalisation techniques.
•To classify messages into positive, negative, or neutral categories using machine learning models such as LSTM.
•To visualise sentiment trends and emotional patterns through graphs and charts.
•To provide insights into communication behaviour and mood dynamics within chat groups or among individuals.







METHODOLOGY

- DATA COLLECTION
- FEATURE EXTRACTING
- MODEL TRAINING
- SENTIMENT ANALYSIS
- VISUALIZATION

CALLENGES

- DATA PRIVACY
- DATA CLEANING COMPLEXITY
- IMBALANCED DATA
- CONTEXT UNDERSTANDING
- PERFORMANCE OPTIMISATION

CONCLUSION

The WhatsApp Chat Sentiment Analyzer successfully demonstrates how NLP and machine learning can extract meaningful emotional insights from chat data. By automating sentiment classification and visualization, the system helps users understand communication patterns, emotional trends, and group dynamics.

RFERENCES

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